

Paper II v1.1 Internal Audit Final Report

Internal author-side audit

21 August 2026 — Overall verdict: PASS

Protocol: INTERNAL_AUDIT_STANDARD_v1.2.

Independence: internal / not external.

Lean execution: not rerun; frozen evidence reviewed only.

The audit covered claims, exact and bounded mathematical checks, static manuscript-to-Lean correspondence, recorded build evidence, bilingual and publication artifacts, prior art and package integrity.

1 Frozen target

English SHA-256: 361802b5398eb9e3cc03b4ca5e591ce7
1987979b02142826054eb5f6a315ded6.
Spanish SHA-256: 26ddd0d08b08e54bfe57e43a9733f456
f94f617e8cfc98e3604b34a3dc10273e.
Lean 4.28.0; Mathlib 8f9d9cff6bd728b17a24e163c9402775d9e6a365.

2 Gate results

Gate	Verdict	Evidence
G0 Target	PASS	6 artifact, 45 source and 62 package hashes; archive sidecar.
G1 Claims	PASS	Exact theorem, branches, degeneracies, scope and quantifiers.
G2 Mathematics	PASS after harness correction	195 cover LPs; $n = 1 \dots 5000$; 139 chordal graphs; 931 copy pairs.
G3 Formal conformance	PASS	Unconditional <code>PaperII.theorem_1_2</code> ; auxiliary surfaces matched.
G4 Recorded build	PASS	8,061 main + 8,032 supplement; 16 axiom outputs; no Lean rerun.
G5 Bilingual	PASS	Headings, formulas, 3 figures, tables and 23 identifiers aligned.
G6 TeX/PDF	PASS	24/24 pages, embedded fonts, full visual inspection.

G7 Prior art	PASS internal	No earlier same exact finite functional/extremizer result identified.
G8 Package	PASS	Active/superseded and evidence structure consistent.

3 Harness correction

The first G2 attempt over-scoped a saturated-branch argmax statement to include the degenerate tie at $n = 2$. The harness domain was corrected and the failed attempt retained. This was a test-specification correction, not a manuscript or mathematical correction. No manuscript artifact required modification.

4 Prior art and external gates

No earlier exact maximum of the fractional functional over chordal graphs was identified internally; the integral Erdős #81 problem remains open. Specialist prior-art review, independent rebuild, external adversarial audit, peer review, tag and release approval remain open.

5 Overall verdict

PASS. All nine blocking internal gates pass. There are zero unresolved blocker or major findings.